

Lab0

Reflection Question 1

1. A socket is a communication point that allows two devices or programs to send and receive data over a network. It is commonly used for communication between a client and a server.
2. AF_INET means that the socket will use IPv4 addresses for network communication.
3. SOCK_STREAM means that the socket will use TCP. TCP provides a reliable connection where data is delivered in the correct order.

```
PS C:\Users\suraj\Desktop\8561> py .\server.py
Server is waiting for a connection...
```

```
PS C:\Users\suraj\Desktop\8561> py .\client.py
Server replied: Server received:
PS C:\Users\suraj\Desktop\8561>
```

```
PS C:\Users\suraj\Desktop\8561> py .\server.py
Server is waiting for a connection...
Connected by: ('127.0.0.1', 51017)
Client says: Hello Server
PS C:\Users\suraj\Desktop\8561>
```

```
PS C:\Users\suraj\Desktop\8561> py .\server.py
Server is waiting for a connection...
```

```
PS C:\Users\suraj\Desktop\8561> py .\client.py
Server replied: Server received: Hello Server this is suraj
PS C:\Users\suraj\Desktop\8561>
```

```
PS C:\Users\suraj\Desktop\8561> py .\server.py
Server is waiting for a connection...
Connected by: ('127.0.0.1', 53257)
Client says: Hello Server this is suraj
PS C:\Users\suraj\Desktop\8561> █
```

```
PS C:\Users\suraj\Desktop\8561> py .\http_client.py
HTTP/1.1 200 OK
Date: Thu, 10 Sep 2026 17:09:07 GMT
Content-Type: text/html
Transfer-Encoding: chunked
Connection: close
Server: cloudflare
Last-Modified: Wed, 02 Sep 2026 22:14:26 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 14204
cf-cache-status: HIT
CF-RAY: a39009e06ec64d45-YVR

22f
<!doctype html><html lang="en"><head><title>Example Domain</title><link rel="icon" href="data:,"><meta name="viewport" content="width=device-width, initial-scale=1"><style>body{background:#eee;width:60vw;margin:15vh auto;font-family:system-ui,sans-serif}h1{font-size:1.5em}div{opacity:0.8}a:link,a:visited{color:#348}</style></head><body><div><h1>Example Domain</h1><p>This domain is for use in documentation examples without needing permission. Avoid use in operations.</p><p><a href="https://iana.org/domains/example">Learn more</a></p></div></body></html>

0

PS C:\Users\suraj\Desktop\8561> █
```

Reflection Question 2

1. The client connected to the host or IP address that was specified in the client program.
2. The client used port 80.
3. Port 80 is relevant because it is the standard port commonly used for HTTP web traffic.
4. The server sent a response back to the client after the connection was established.

```
PS C:\Users\suraj\Desktop\8561> python socketio_server.py
Socket.IO server running...
===== Running on http://0.0.0.0:8080 =====
(Press CTRL+C to quit)
█
```

```
PS C:\Users\suraj\Desktop\8561> python socketio_client.py
websocket-client package not installed, only polling transport is available
Connection established
Sending message: Hello Server
Disconnected from server
□
```

```
PS C:\Users\suraj\Desktop\8561> python socketio_server.py
Socket.IO server running...
===== Running on http://0.0.0.0:8080 =====
(Press CTRL+C to quit)
Client connected: tOsBWdrAjNC2K0maAAAB
Message received from client: Hello Server
Client disconnected: tOsBWdrAjNC2K0maAAAB
□
```

Part 8 - Security Reflection

1. If a server accepts connections from anyone, unauthorized users could connect to it. They could send malicious requests, abuse the service, or attempt to exploit vulnerabilities.
2. No, the Lab 0 message is not encrypted. Since it uses a basic TCP socket without encryption such as TLS, the message is sent as plaintext.
3. Yes, network traffic could potentially be observed by another system. Since the traffic is not encrypted, someone who is able to capture the network traffic could potentially read the data being transmitted.
4. Servers should validate input to make sure they only process expected and safe data. Without input validation, malicious or unexpected data could cause errors or potentially be used to exploit the application.
5. Authentication is useful because it allows the server to verify who is connecting. This helps prevent unauthorized users from accessing the server.
6. Two ways to make the application more secure would be to encrypt the communication using TLS and add authentication so that only authorized users or clients can connect.

What I learned:

In this lab, I learned how to create and run Python programs, work with TCP sockets, and build a basic client-server application using both the socket library and Socket.IO. I also learned how a client connects to a server, sends messages, and receives responses. The hardest part for me was setting up GitHub and correctly pushing my files to the repository because I had some issues with the branches and Git commands. After troubleshooting those problems, I gained a better understanding of how Git and GitHub are used to manage and submit code.

<https://github.com/surajnand/FSC8561-Labs-Suraj/tree/Lab0>

